

Technical Document #451: Retrieval Augmented Generation - Comprehensive Overview

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Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Chunking splits documents into smaller segments for embedding.
- Vector databases store dense embeddings for semantic search.